

Position and Velocity of SHM

$$x = A \sin(\omega t + \Phi_0)$$

$$v = A \omega \cos(\omega t + \Phi_0)$$

$$v = \omega \sqrt{A^2 - x^2}$$

$$\text{a) } x=0 \rightarrow v_{\max} = A\omega$$

$$\text{b) } x = \frac{A}{2} \rightarrow v = \frac{\sqrt{3}A}{2}\omega$$

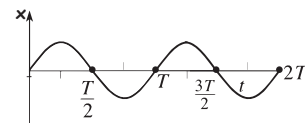
$$\text{c) } x = \frac{A}{\sqrt{2}} \rightarrow v = \frac{A\omega}{\sqrt{2}}$$

$$\text{d) } x = A \rightarrow v = 0$$

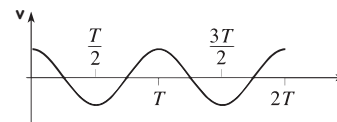
Graphical Representation of Position and Velocity

Start from mean position, $\Phi_0 = 0$

$$x = A \sin(\omega t)$$

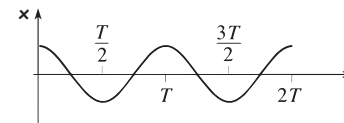


$$v = A\omega \cos(\omega t)$$

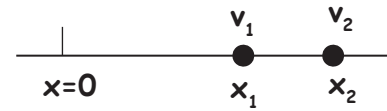
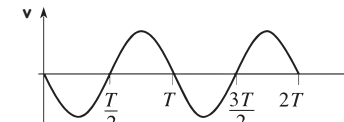


Start from extreme position $\Phi_0 = \frac{\pi}{2}$

$$x = A \cos(\omega t)$$



$$v = -A\omega \sin(\omega t)$$



$$A = \sqrt{\frac{v_1^2 x_2^2 - v_2^2 x_1^2}{v_1^2 - v_2^2}}$$

$$\omega = \sqrt{\frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}}$$

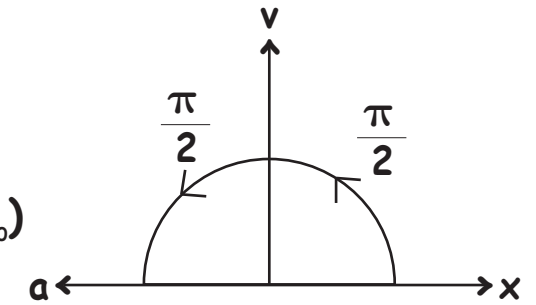
Acceleration of SHM

$$x = A \sin(\omega t + \Phi_0)$$

$$v = A\omega \cos(\omega t + \Phi_0)$$

$$a = -A\omega^2 \sin(\omega t + \Phi_0)$$

$$a_{\max} = -A\omega^2$$



Phase difference between x and $v = \frac{\pi}{2}$

Phase difference between v and $a = \frac{\pi}{2}$

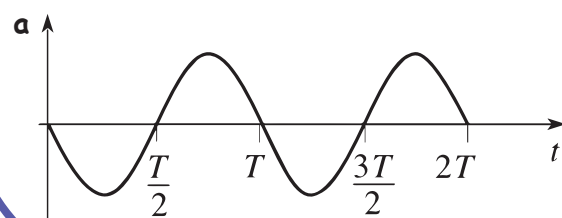
Phase difference between x and $a = \pi$

OSCILLATIONS 01

Graphical Representation of Acceleration

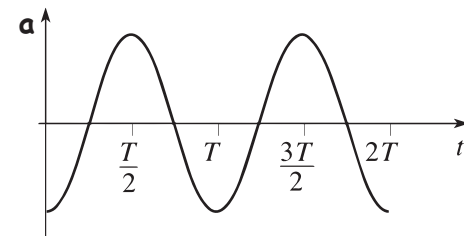
Start from mean position, $\Phi_0 = 0$

$$a = -A\omega^2 \sin(\omega t)$$



Start from extreme position, $\Phi_0 = \frac{\pi}{2}$

$$a = -A\omega^2 \cos(\omega t)$$



Calculation of Time period and amplitude

$$v_{\max} = A\omega$$

$$a_{\max} = A\omega^2$$

$$\omega = \frac{a_{\max}}{v_{\max}}$$

$$A = \frac{v_{\max}^2}{a_{\max}}$$

$$\frac{2\pi}{T} = \frac{a_{\max}}{v_{\max}}$$

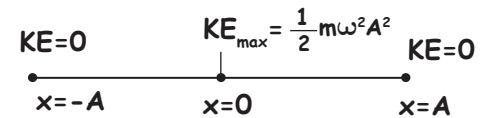
$$T = 2\pi \frac{v_{\max}}{a_{\max}}$$



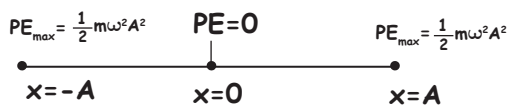
**PHYSICS
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Energy of SHM

$$K.E = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - x^2)$$



$$P.E = \frac{1}{2}m\omega^2 x^2$$



$$\text{Total mechanical energy, } E = \frac{1}{2}m\omega^2 A^2 = \text{Constant}$$

$$1) x = \frac{A}{\sqrt{2}}$$

$$K.E = P.E = \frac{E}{2}$$

$$2) x = \frac{A}{2}$$

$$K.E = \frac{3E}{4}, P.E = \frac{E}{4}$$

$$3) x = 0$$

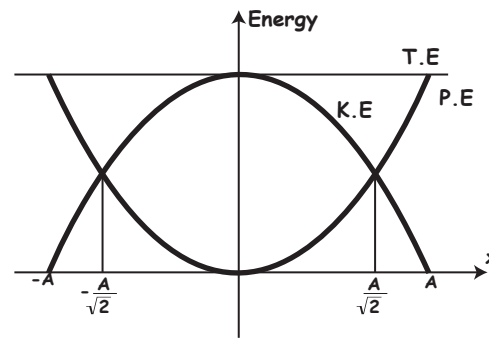
$$K.E = E$$

$$P.E = 0$$

$$4) x = A$$

$$K.E = 0$$

$$P.E = E$$



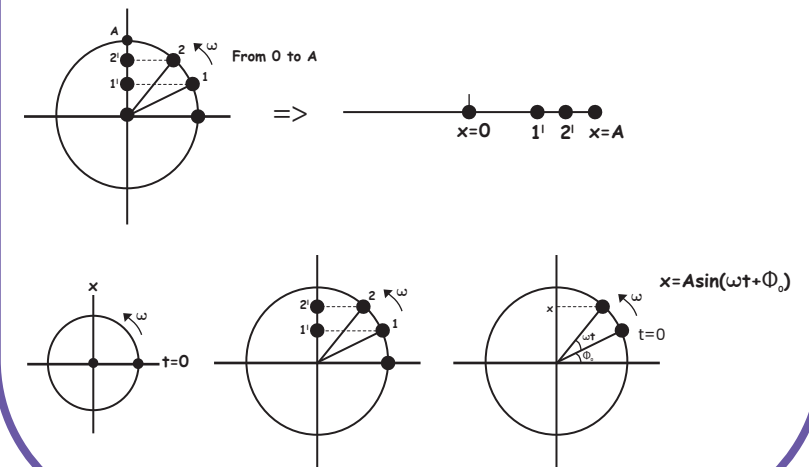
$$\text{Total mechanical energy (E)} = \frac{1}{2}m\omega^2 A^2$$

Note: In SHM, if particle oscillates with frequency ω , then the K.E & P.E oscillate with 2ω

OSCILLATIONS 02

PROJECTION OF CIRCULAR MOTION

Projection/shadow of uniform circular motion on y axis is SHM



Two particles executing SHM meet at $x = \frac{\sqrt{3}A}{2}$

for particle A

$$x = \frac{\sqrt{3}A}{2}, t=0 \Rightarrow \frac{\sqrt{3}A}{2} = A \sin \phi_A$$

$$\phi_A = 60^\circ = \frac{\pi}{3}$$

for particle B

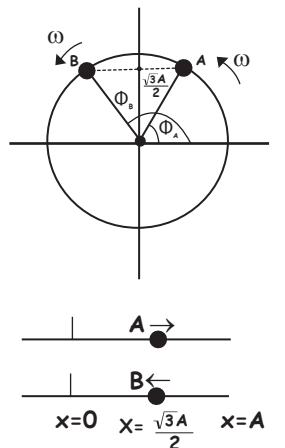
$$x = \frac{\sqrt{3}A}{2}, t=0 \Rightarrow \frac{\sqrt{3}A}{2} = A \sin \phi_B$$

$$\phi_B = 90^\circ + 30^\circ$$

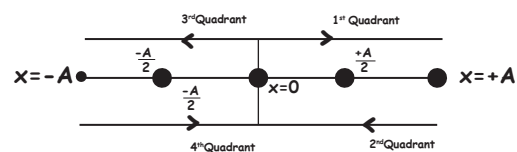
$$\frac{\pi}{2} + \frac{\pi}{6} = \frac{2\pi}{3}$$

Phase difference between particles

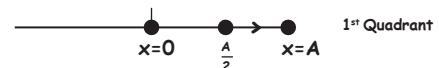
$$\phi = \frac{2\pi}{3} - \frac{\pi}{3} = \frac{\pi}{3} = 60^\circ$$



INITIAL PHASE FROM POSITION & DIRECTION



Eg: Particle is at $x = \frac{A}{2}$ [$t=0$] and moves towards A



1st Quadrant

1) Particle is at $x = \frac{A}{2}$ [$t=0$] and moves towards A

$$x = \frac{A}{2}, t=0 \Rightarrow \frac{A}{2} = A \sin \phi_0$$

$$\phi_0 = 30^\circ = \frac{\pi}{6}$$

$$x = A \sin(\omega t + \frac{\pi}{6})$$

2) Particle is at $x = \frac{A}{\sqrt{2}}$ [$t=0$] and moves towards A

$$x = \frac{A}{\sqrt{2}}, t=0 \Rightarrow \frac{A}{\sqrt{2}} = A \sin \phi_0$$

$$\phi_0 = 45^\circ = \frac{\pi}{4}$$

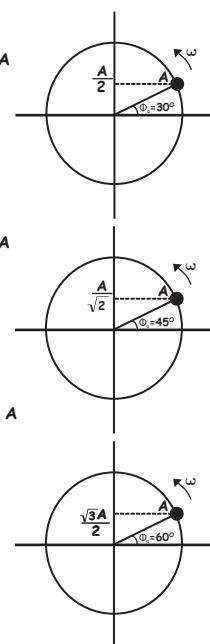
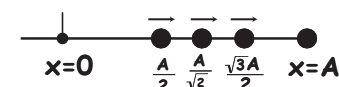
$$x = A \sin(\omega t + \frac{\pi}{4})$$

3) Particle is at $x = \frac{\sqrt{3}A}{2}$ [$t=0$] and moves towards A

$$x = \frac{\sqrt{3}A}{2}, t=0 \Rightarrow \frac{\sqrt{3}A}{2} = A \sin \phi_0$$

$$\phi_0 = 60^\circ = \frac{\pi}{3}$$

$$x = A \sin(\omega t + \frac{\pi}{3})$$



2nd Quadrant

1) Particle is at $x = \frac{A}{2}$ [$t=0$] and moves towards O

$$x = \frac{A}{2}, t=0 \Rightarrow \frac{A}{2} = A \cos \phi_0$$

$$\phi_0 = 90^\circ - 60^\circ = \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}$$

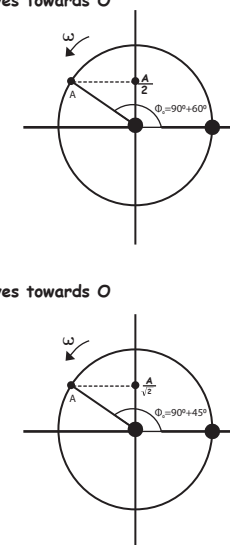
$$x = A \cos(\omega t + \frac{\pi}{6})$$

2) Particle is at $x = \frac{A}{\sqrt{2}}$ [$t=0$] and moves towards O

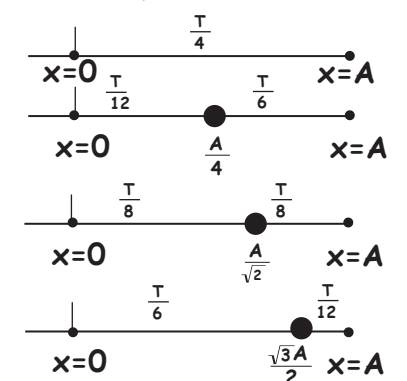
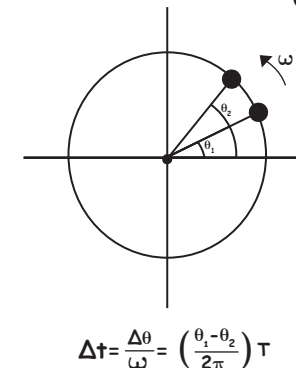
$$x = \frac{A}{\sqrt{2}}, t=0 \Rightarrow \frac{A}{\sqrt{2}} = A \cos \phi_0$$

$$\phi_0 = 90^\circ - 45^\circ = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

$$x = A \cos(\omega t + \frac{\pi}{4})$$

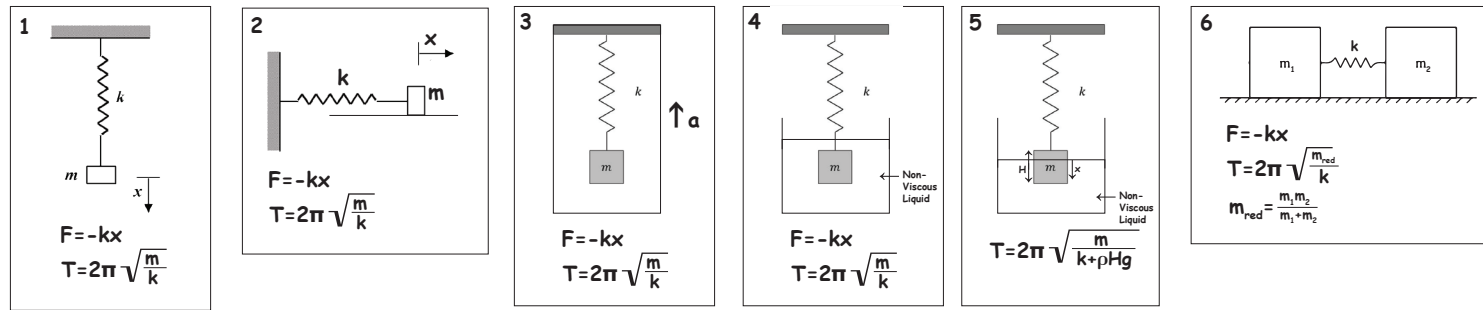


Calculation of time

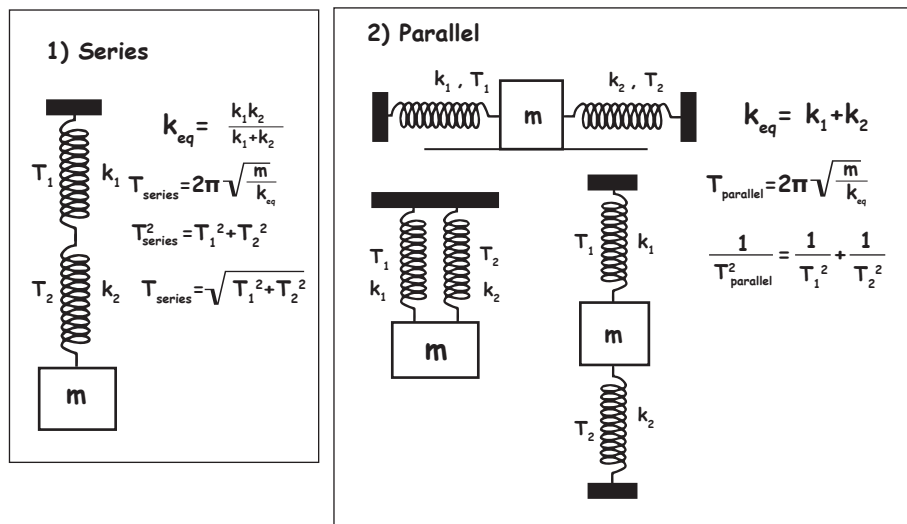


OSCILLATIONS 03

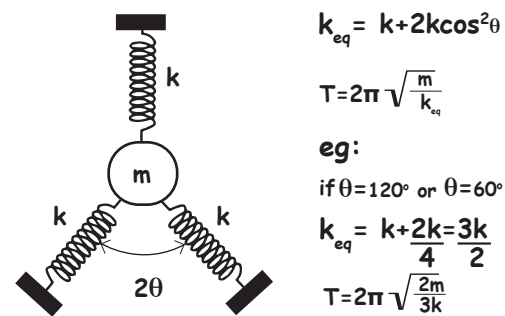
TIME PERIOD OF S.H.M



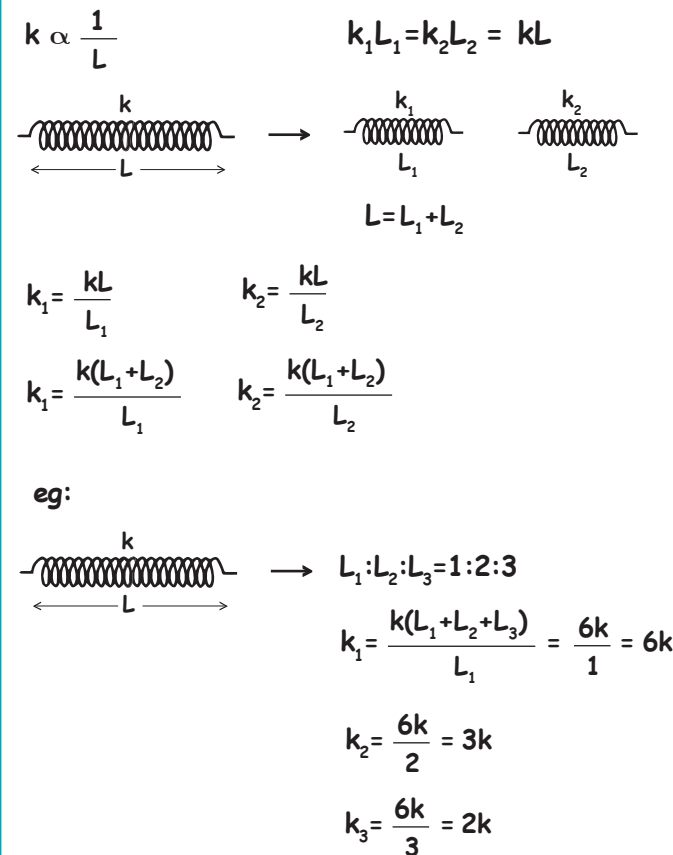
COMBINATIONS OF SPRINGS



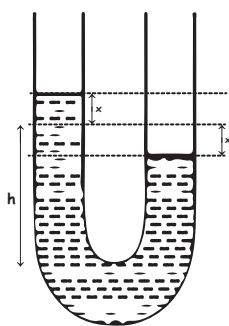
SPECIAL CASE



CUTTING OF SPRINGS



OSCILLATION OF LIQUID COLUMN

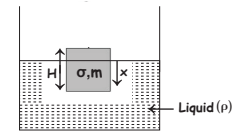


$$T = 2\pi \sqrt{\frac{m}{2\rho gA}}$$

$$T = 2\pi \sqrt{\frac{h}{g}}$$

m = mass of liquid
 ρ = Density of liquid
 A = Area of U-tube

OSCILLATION OF FLOATING BODY

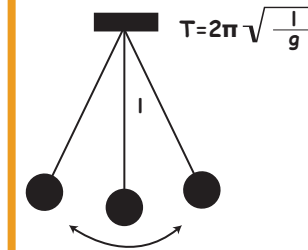


$$T = 2\pi \sqrt{\frac{m}{\rho gA}}$$

$$T = 2\pi \sqrt{\frac{\sigma H}{\rho g}}$$

m = mass of body
 σ = Density of Body
 ρ = Density of liquid

SIMPLE PENDULUM



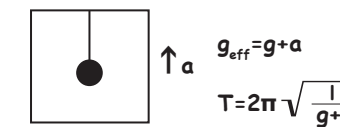
• Second's pendulum
 $T = 2$ second
 $l = 1$ meter

Concept of $g_{effective}$

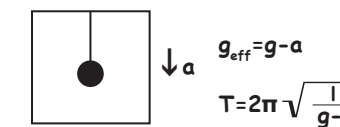
$$T = 2\pi \sqrt{\frac{l}{g_{eff}}}$$

Pendulum in lift

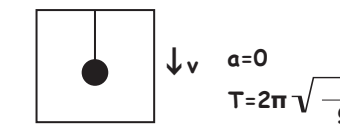
Case 1-Moving with constant upward acceleration 'a'



Case 2-Moving with constant downward acceleration 'a'



Case 3-Moving with constant velocity

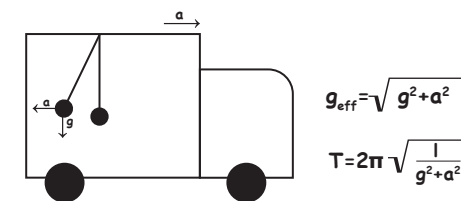


Case 4-Free fall

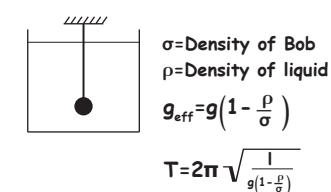
$a = g$

$$g_{eff} = g - a = g - g = 0 \quad T \rightarrow \infty$$

Pendulum in a truck moving with constant acceleration



Pendulum in Water



DIFFERENTIAL EQUATION OF S.H.M

$$ma = -kx$$

$$m \frac{d^2x}{dt^2} = -kx$$

$$\frac{d^2x}{dt^2} = \frac{-k}{m} x$$

$$\frac{d^2x}{dt^2} + \frac{k}{m} x = 0$$

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

where, $\omega^2 = \frac{k}{m}$ $\omega = \sqrt{\frac{k}{m}}$

Solving,
 $x = A \sin(\omega t + \phi_0)$

A = Amplitude of SHM

ϕ_0 = Initial phase angle

Eg: $\left[\begin{array}{l} m = 4 \text{ Kg} \\ K = 320 \text{ N/m} \end{array} \right]$

$$4 \frac{d^2x}{dt^2} + 320 x = 0$$

$$\frac{d^2x}{dt^2} + 80 x = 0$$

$$\omega^2 = 80 \quad \omega = \sqrt{80}$$

$$\frac{2\pi}{T} = \sqrt{80}$$

$$T = \frac{2\pi}{\sqrt{80}}$$



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